Manav Mahesh Patel

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Summary

Mechanical Engineer with expertise in HVAC layout, MEP coordination, and pipe sizing using **Revit MEP**, **AutoCAD**, and **ANSYS**. Proficient in **ASME B31** standards, technical documentation (MTOs/Bids), and thermal-structural analysis. Eager to apply collaborative design skills in process piping and plumbing as a mechanical engineering intern.

Featured Achievement

Patent – Pedal Box Assembly for Electric Vehicles | *Design Patent No. 471278-001* **2024**
- Co-invented a patented modular pedal box assembly; applied design-for-manufacturing (DFM) and ergonomic principles to optimize structural integrity and user interface.

Education

University of Colorado Boulder **Boulder, CO**
Master of Science in Mechanical Engineering GPA : 3.944/4.0 *August 2025 – Present*
- Relevant Courses: Fluid Dynamics, Heat Transfer, Thermal System Innovation, Advanced Product Design

Dwarkadas J Sanghvi College of Engineering **Mumbai, India**
Bachelor of Technology in Mechanical Engineering GPA: 3.4/4.0 *July 2020 – May 2024*
- Relevant Courses: Refrigeration and Air Conditioning, FEA, CFD, Machine Design, Thermodynamics, Fluid Mechanics

Technical Skills

HVAC & Piping Design: Revit MEP, **AutoPipe**, HVAC System Layout, Ductwork Design, Pipe Sizing, Equipment Scheduling, Chilled Water Systems, ASME B31.1/B31.3, Piping Isometrics, MTOs

CAD / BIM Software: AutoCAD, SolidWorks, **Revit (MEP/Systems)**, Inventor CAD

Analysis Tools: ANSYS Mechanical (FEA, Thermal), MATLAB, Microsoft Excel (Advanced) , Microsoft Office

Engineering Standards: ASME B31, ASHRAE, HVAC Load Calculations, Piping Stress Analysis, Pressure Vessel Fundamentals

Business & Support: Tendering & Bid Preparation, Technical Documentation, BOM Management, Project Coordination

Professional Experience

Star Pipe Products **Houston, TX**
CAD Design Intern *June 2026 – Present*
- Engineered comprehensive 3D CAD models, component parts, and designs for work holding equipment using **SolidWorks** to support manufacturing workflows. Using Parametric Design tables to create uniformly usable components.
- Mapped manufacturing warehouse facility spaces and designed accurate architectural machine pits and structural footprints required to house heavy industrial machinery.
- Generated detailed 2D manufacturing drawings, shop floor safety and assembly documentation to guide cross-functional plant floor safety and optimize workflow.

M/s. ASMACS **Remote / Field**
Junior Engineer – Tendering & Coordination *Nov 2024 – June 2025*
- Conducted technical research and prepared comprehensive bid and tender packages for high-profile process piping projects for clients including the Indian Navy, Mazagon Dockyard Limited, and HAL.
- Coordinated between site engineers and office leadership to resolve technical issues in layouts and systems installations, maintaining strict project schedules.
- Maintained and organized critical engineering deliverables including isometric drawings, equipment datasheets, and material take-offs (MTOs) and manpower.

DJS Racing – SAE Team **Mumbai, India**
Senior Member – Brakes & Structural Systems *June 2022 – May 2024*
- **Led design, structural stress, and thermal-fluid analysis** for a Sliding Brake Pedal Assembly and brake rotors using **ANSYS Mechanical** and **Fluent**.
- Optimized component performance under thermal and mechanical loading using FEA principles.
- Communicated technical design decisions within a 50+ member multidisciplinary team.

Projects

L.A.M.P. — Modular Lever Arm Assist | *SolidWorks, ANSYS, DFM, A356 Aluminum* **Jan 2026 – May 2026**
- Engineered a modular retrofit attachment for material handling equipment, utilizing **SolidWorks** and **3D printing** to develop prototypes that improved mechanical leverage and reduced user physical strain.
- Conducted first-order structural analysis modeling the assembly as a cantilever beam to evaluate bending moments and stress, ensuring the design safely sustained operational loads up to **150 lbs**.
- Developed a universal V-clamp interface and performed a **DFM/Business analysis** to select a hybrid sand-cast **A356 Aluminum** and steel tube construction, optimizing the product's strength-to-weight ratio.

Certifications

Revit & BIM: Essential Training for MEP, Sustainable HVAC Systems with Revit, Sprinkler Design
Piping & Mechanical: ASME B31 Piping Engineering Masterclass, HVAC/Plumbing Drawing Reading